

SAMUEL NDUBUISI

(330) 396-2395 | samuelndubuisi32@gmail.com | [linkedin.com/in/samuel-ndubuisi-a4792a220](https://www.linkedin.com/in/samuel-ndubuisi-a4792a220) | github.com/Sammieblz | smdubuisi.vercel.app

PROFESSIONAL SUMMARY

Software engineer with 4+ years across the full stack, from customer-facing web products and payment systems to machine vision pipelines and cloud infrastructure. Comfortable moving between React and C++, between a CI/CD pipeline and a PyTorch training loop. Recently won two tracks at Kent Hack Enough 2026, building a voice-first AI interview coach.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C, C++, C#, Java, Bash/Shell

Frontend: React, Next.js, Vue.js, Quasar, Svelte, Angular, HTML, CSS

Backend & APIs: Node.js, .NET, REST, gRPC, SOAP, GraphQL, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, SQL

DevOps & Cloud: Docker, Kubernetes, Azure DevOps, Jenkins, GitHub Actions, Git, TFS, SVN, Terraform, Ansible, AWS (EC2, S3, Lambda, API Gateway, Amplify), Azure (App Services, Functions, API Management)

ML/CV & AI: OpenCV (real-time video pipelines, Haar cascades), PyTorch, prompt engineering, voice and real-time audio pipelines (ElevenLabs WebSockets), Linux (RHEL, Ubuntu, Debian), TCP/IP, DNS, TLS, SSH/SFTP, CDN, Load Balancing

Agentic AI Coding: Claude (Anthropic), GitHub Copilot; RAG architecture design and implementation

Embedded (Project-Based): Yocto Project, BitBake, systemd, cross-compilation, QEMU, Modbus/TCP

PROFESSIONAL EXPERIENCE

Software Engineer | Antares Vision Group (Rigid Containers)

Jan 2025 - Present

- Maintained and enhanced machine vision algorithms using OpenCV and physics-based models to improve defect detection in rigid-container inspection.
- Developed and trained PyTorch-based defect classification models, reducing manual inspection time by 25% in production environments.
- Collaborated with firmware teams to implement a multithreaded inspection system, improving real-time throughput by 30%.
- Maintained and debugged legacy enterprise software using MFC/C++, PostgreSQL, and gRPC to ensure reliability across global deployments.
- Designed and maintained a microservice architecture using .NET (C#), C++, and Vue.js, with gRPC for inter-service communication and REST APIs for backend-frontend integration.
- Contributed to migrating a legacy system to Vue.js/Quasar, .NET (C#), PostgreSQL, gRPC APIs, and Consul; prototype available for testing in Mar 2025.
- Wrote unit and integration tests (GoogleTest, Jest/Vitest) and improved delivery with Azure DevOps pipelines, SonarCloud quality gates, Docker, and Kubernetes.

Freelance Software Developer | Fiverr / Upwork / Byteflow

Jan 2022 - Nov 2025

- Designed and delivered API-driven features tailored to client requirements to improve real-time functionality and user experience.
- Developed a Java/Kotlin mobile application enabling personalized UX and more efficient data management.
- Built Buckeye Bin Cleaning: a Next.js customer booking web app with a secure admin dashboard for scheduling and business operations.
- Built course delivery system: implemented Stripe subscriptions and billing logic for a curriculum-based learning platform serving 2,500 users at launch and growing.
- Migrated a legacy .NET web app to a Contentful CMS-driven model/component rendering architecture; generated Contentful model types via GraphQL and built a reusable React rendering registry.
- Leveraged AWS services (Amplify, EC2) to establish CI/CD from repository to production builds with automated test execution.

Application Engineer Intern - IS Infrastructure | Snap-on Business Solutions

Dec 2022 - Aug 2024

- Managed backend infrastructure and automated routine tasks using Python and Ansible, resulting in a 15% reduction in system downtime.
- Engineered AWS deployment workflows using Terraform, Ansible, Git, and Jenkins; scaled Linux server clusters and improved uptime by 20%.
- Helped migrate an internal Java-based legacy system to TypeScript, React, and Node.js, improving maintainability and scalability.
- Delivered software solutions using Scrum and Jira to keep projects aligned with business goals and deadlines.

EDUCATION

M.S., Software Engineering (In Progress) - University of Arizona

Nov 2025 - Aug 2027 (Expected)

B.S., Computer Information Systems - Software Development - University of Akron

Aug 2019 - Dec 2024

A.A.B., Applied Business in Computer Information Systems - University of Akron

May 2024

Honors: Dean's List (Fall 2022, Spring 2023)

SELECTED PROJECTS

InterviewAI ([Devpost](#) | [GitHub: Sammieblz/AITI](#)) | Kent Hack Enough 2026: Winner - Data Science Track; Winner - MLH Best Use of ElevenLabs

- Built a voice-first behavioral interview coach that runs mock interviews, scores answers, and provides actionable STAR-based feedback.

- Owned the full stack (Next.js/React, Python, AWS) and ML pipeline: built the complete training workflow from behavioral dataset preparation and synthetic corpus generation to PyTorch fine-tuning, producing independent model checkpoints importable by backend services; integrated ElevenLabs WebSockets for low-latency voice output.
- Drove coaching quality through prompt engineering: role-specific question generation, STAR-based scoring, and feedback that distinguishes individual from team contributions.
- Hardened real-time audio UX across browsers, including mic permissions, reconnect/retry logic, and recovery from dropped WebSocket sessions.

VisionPlayground (github.com/Sammieblz/VisionPlayground)

- Smart AI dashcam and driver monitoring prototypes with real-time eye-blink and eye-closure detection from a live camera feed using OpenCV (Haar cascade eye detector).
- Implemented consecutive-frame thresholding and event triggers (PyAutoGUI) as a proof-of-concept for alerting and recording actions.

Brack (github.com/Sammieblz/brack-app)

- Digital book tracking app that lets users manage reading lists, track progress, and log reading activity across sessions.
- Integrated a built-in book recommendation model that surfaces personalized title suggestions based on reading history and preferences.

Material UI (github.com/Sammieblz/material-ui) | Open Source Contributor

- Contributing to Material UI, Google's Material Design React component library used by millions of developers worldwide.
- Work spans component enhancements and fixes across the React component ecosystem, engaging with one of the most widely adopted open-source UI libraries in the JavaScript community.